

Clifford scalar potentials and Dirac confinement in every dimension

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Result. The Clifford-algebra problem in Comment 5 of Nenciu–Nenciu–Obermeyer is solved for the minimal Hamiltonian spinor in every dimension. The scalar-potential space has real dimension one for even spatial dimension and two for odd spatial dimension. The source confinement theory extends to this full class, and the fixed-mass boundary threshold $|\lambda| = 1/2$ is sharp in every dimension.

1 Source question

Nenciu, Nenciu, and Obermeyer [1] define a Hermitian matrix potential V to be *scalar* when it anticommutes with every principal Dirac matrix. Their Comment 5 asks for free Dirac matrices in all dimensions, the complete class $\mathcal{S}_s$ of such potentials, and the analogue of their Lorentz-scalar theory. They separately ask whether the optimal boundary growth rate is dimension-independent. The question is on PDF pages 34–35.

$$\left\{ \sum_{\ell=1}^{16} v_{\ell}(x) \Gamma_{\ell}, \alpha_j \right\} = 0 \quad \text{for all } j = 1, 2, 3.$$

While we do not find here the general form of a scalar potential, (7.2) implies that $v_{12}\Gamma_{12} \in \mathcal{S}_s$, which in turn yields that $\mathcal{S}_{Ls}$ is a strict subset of $\mathcal{S}_s$. In addition, (7.2) also shows that the set of matrices $\{\alpha_1, \alpha_2, \alpha_3, \Gamma_{12}\}$ satisfies the same anticommutation relations (4.4) as the set $\{\alpha_1, \alpha_2, \alpha_3, \beta\}$. Since the proofs in Section 4 only use these anticommutation relations (and not the concrete form of the matrices involved), we conclude that the criteria from Section 4 apply to the operator $\mathbb{D}_0 + v_{12}\Gamma_{12}$. Turning to dimension $d = 4$, one can consider the following

$$\mathbb{D}_0 = \sum_{j=1}^3 \alpha_j D_j + \Gamma_{12} D_4$$

as the free Dirac operator, and note that, by (7.2), $v_2\beta \in \mathcal{S}_s$, meaning $\mathcal{S}_s \neq \emptyset$.

Based on the discussion above, we formulate the following:

Open problem: Using the Clifford algebra formalism, write down the free Dirac operators in all dimensions d , identify the class of scalar potentials $\mathcal{S}_s$, and develop the analogous theory to that of Section 4

One final note: In the cases considered in Section 4 the optimal increase rate at the boundary of Ω for Lorentz scalar potentials is independent of the dimension d . It is a natural question whether this is a generally valid fact in all dimensions. First recall that this is indeed the case for Schrödinger operators (see [32]). At the technical level, one reason for this is the fact that the best constant in the Hardy inequality is dimension-independent, and the same Hardy inequality enters our proofs in Section 4. Heuristically, the loss of essential self-adjointness is caused by the fact that the particle reaches the boundary $\partial\Omega$, and hence only the motion along the normal to $\partial\Omega$ matters. In addition, recall that, in many respects, Lorentz scalar potentials behave as positive potentials in the Schrödinger case

Figure 1. The algebraic problem and the dimension-independence question, arXiv:2010.09816, PDF page 34.

(see e.g. [30, 40]). All of the above make it tempting to think that the optimal increase rate is indeed dimension-independent.

6. We now consider the case of the Dirac operator with an electric potential (see [42] Ch. 4.2.2)), that is

Figure 2. The final two lines of the dimension-independence question, PDF page 35.

2 Explicit all-dimensional model

Let $d \geq 1$ and $N_d = 2^{\lfloor (d+1)/2 \rfloor}$. Write X, Y, Z for the Pauli matrices. The standard Jordan–Wigner matrices give an explicit irreducible complex representation of Cl_{d+1} : if $m = \lfloor (d+1)/2 \rfloor$, put

$$\begin{aligned}\gamma_{2j-1} &= Z^{\otimes(j-1)} \otimes X \otimes \mathbf{l}^{\otimes(m-j)}, \\ \gamma_{2j} &= Z^{\otimes(j-1)} \otimes Y \otimes \mathbf{l}^{\otimes(m-j)}, \quad 1 \leq j \leq m,\end{aligned}$$

retaining only indices at most $d+1$; when $d+1 = 2m+1$, append $\gamma_{2m+1} = Z^{\otimes m}$. Thus

$$\gamma_j^* = \gamma_j, \quad \gamma_j \gamma_k + \gamma_k \gamma_j = 2\delta_{jk} \mathbf{l}.$$

Set $\alpha_j = \gamma_j$ for $1 \leq j \leq d$, $\beta = \gamma_{d+1}$, and

$$\mathbf{D}_0 = -i \sum_{j=1}^d \alpha_j \partial_j, \quad \Gamma_d = i^{d(d-1)/2} \alpha_1 \cdots \alpha_d.$$

The phase makes Γ_d a Hermitian involution. It anticommutes with each α_j for even d and commutes with each α_j for odd d .

3 Proof intuition

Multiplication by the already available mass matrix β turns an anticommuting matrix into a commuting one. The problem is therefore exactly a commutant calculation for the restriction from Cl_{d+1} to Cl_d . That restriction is irreducible in even d , but splits into the two inequivalent irreducibles in odd d ; the volume element Γ_d distinguishes the two summands. This parity fact produces either one or two real mass directions.

Once these directions are known, their Clifford relations collapse V^2 to the Euclidean squared length of the coefficient vector. The source paper’s coercivity matrix can then be bounded by a scalar expression. For a fixed mass direction, the critical proof is literally dimension-free: the matrices $iM\alpha_j$ again form a Clifford family and $\|\mathbf{D}_0\phi\|_2^2 = \|\nabla\phi\|_2^2$. On the unit ball the normal equation has exponents $\pm\lambda$, so square integrability changes precisely at $|\lambda| = 1/2$.

4 Classification theorem

For a point x , let

$$\mathcal{S}_s(x) = \{V = V^* \in M_{N_d}(\mathbb{C}) : \alpha_j V + V \alpha_j = 0 \text{ for } j = 1, \dots, d\}.$$

Theorem 1 (Complete scalar-potential classification). *For the minimal Hamiltonian spinor model above,*

$$\mathcal{S}_s(x) = \begin{cases} \{v(x)\beta : v(x) \in \mathbb{R}\}, & d \text{ even}, \\ \{v_1(x)\beta + v_2(x)\tilde{\beta} : v_1(x), v_2(x) \in \mathbb{R}\}, & d \text{ odd}, \end{cases} \quad \tilde{\beta} = i\beta\Gamma_d.$$

For odd d , β and $\tilde{\beta}$ are anticommuting Hermitian involutions. Consequently

$$(v_1\beta + v_2\tilde{\beta})^2 = (v_1^2 + v_2^2)\mathbf{l}.$$

In the standard $d = 3$ matrices of [1], $\tilde{\beta} = -\Gamma_{12}$, up to the harmless choice of orientation. Hence the two matrices displayed in Comment 5 span the entire class.

Proof. Let V anticommute with all α_j and put $C = \beta V$. Twice using anticommutation gives $\alpha_j C = C \alpha_j$, so C belongs to the commutant of the restricted Cl_d representation.

If d is even, complex Cl_d is a full matrix algebra and its action on $\mathbb{C}^{N_d}$ is irreducible. Schur's lemma gives $C = c\text{I}$, hence $V = c\beta$; Hermiticity forces $c \in \mathbb{R}$.

If d is odd, the restriction is the multiplicity-one sum of the two inequivalent irreducible Cl_d modules. The two eigenspaces of the central volume element Γ_d are precisely those summands. Therefore its commutant is $\text{span}_{\mathbb{C}}\{\text{I}, \Gamma_d\}$. Thus $V = a\beta + c\beta\Gamma_d$. Since $\beta\Gamma_d = -\Gamma_d\beta$ is skew-Hermitian, $V = V^*$ holds exactly when $a \in \mathbb{R}$ and $c = ib$ with $b \in \mathbb{R}$. This proves the displayed classification.

For odd d , β anticommutes with Γ_d . Direct multiplication now shows that $i\beta\Gamma_d$ is Hermitian, squares to I , and anticommutes with β and every α_j . The square formula follows. Substitution of the standard $d = 3$ matrices gives $i\beta\Gamma_3 = -\Gamma_{12}$. $\square$

5 The full mass-vector confinement criterion

Let $q_d = 1$ in even dimensions and $q_d = 2$ in odd dimensions. Denote the mass frame by $B_1 = \beta$, and, in odd dimensions, $B_2 = \tilde{\beta}$. For $u = (u_1, \dots, u_{q_d}) \in C^1(\Omega; \mathbb{R}^{q_d})$, set $V_u = \sum_a u_a B_a$. Let $\hat{\delta}$ be the regularized distance used in [1, Theorem 2.2] and put $h = \log \hat{\delta}$.

Theorem 2 (All-dimensional scalar criterion). *The source coercivity matrix [1, Theorem 2.5] specializes exactly to*

$$Q_{u,h} = |u|^2 \text{I} - i \sum_{j=1}^d \alpha_j \partial_j V_u - 2i\alpha(\nabla h)V_u - |\nabla h|^2 \text{I}. \quad (1)$$

If $Q_{u,h} \geq c\text{I}$ near $\partial\Omega$ for some $c > 0$, then $\text{D}_0 + V_u$ is essentially self-adjoint on $C_c^1(\Omega; \mathbb{C}^{N_d})$. In particular, it is sufficient that

$$|u|^2 - \sqrt{d}|\nabla u|_{\text{F}} - 2|u||\nabla h| - |\nabla h|^2 \geq c \quad (2)$$

near the boundary.

Consequently, let $a > 1$. If for some $\varepsilon > 0$ and positive c, C ,

$$c\delta^{-a} \leq |u| \leq C\delta^{-(2a-1-\varepsilon)}, \quad |\nabla u|_{\text{F}} \leq C\delta^{-(2a-\varepsilon)} \quad (3)$$

when $\delta = \text{dist}(x, \partial\Omega)$ is small, then $\text{D}_0 + V_u$ is essentially self-adjoint.

Proof. Theorem 1 gives $V_u^2 = |u|^2 \text{I}$. Since every $\partial_j V_u$ anticommutes with α_j ,

$$-\frac{i}{2} \sum_j (\alpha_j \partial_j V_u - \partial_j V_u \alpha_j) = -i \sum_j \alpha_j \partial_j V_u.$$

Likewise $-i[\alpha(\nabla h), V_u] = -2i\alpha(\nabla h)V_u$ and $\alpha(\nabla h)^2 = |\nabla h|^2 \text{I}$. This is (1), and the first claim is exactly [1, Theorem 2.5] with the constant Dirac velocity matrix.

Moreover $\|\partial_j V_u\| = |\partial_j u|$ and $\|\alpha(\nabla h)V_u\| = |\nabla h||u|$. The triangle and Cauchy-Schwarz inequalities give (2). Finally $|\nabla h| = O(\delta^{-1})$. Under (3), the negative terms in (2) are respectively $O(\delta^{-2a+\varepsilon})$, $O(\delta^{-2a+\varepsilon})$, and $O(\delta^{-2})$, whereas $|u|^2 \geq c^2\delta^{-2a}$. Since $a > 1$, the positive term dominates. $\square$

6 Critical rate and its sharpness

Theorem 3 (Dimension-free sharp threshold). *Let $\Omega \subset \mathbb{R}^d$ be bounded with C^2 codimension-one boundary, let M be any constant Hermitian involution anticommuting with every α_j , and suppose, near the boundary,*

$$V(x) = \lambda \frac{\ell(x)}{\delta(x)} M, \quad |\ell| \geq 1, \quad |\nabla(\ell/\delta)| \leq C|\ell|/\delta^2.$$

Put $\mu = \limsup_{\delta(x) \rightarrow 0} \delta(x) |\nabla \ell(x)| / |\ell(x)|$. Then $D_0 + V$ is essentially self-adjoint whenever $|\lambda| > (1 + \mu)/2$. Its closed domain is independent of $|\lambda|$ in this range. If Ω is convex and $\ell = 1$ near the boundary, the endpoint $|\lambda| = 1/2$ is included.

For the unit ball B^d , the operator

$$D_0 + \frac{\lambda}{1 - |x|} M \quad \text{on } C_c^1(B^d; \mathbb{C}^{N_d})$$

is essentially self-adjoint if and only if $|\lambda| \geq 1/2$. Thus both the critical growth exponent one and the coefficient $1/2$ are dimension-independent and optimal.

Proof. The proof of [1, Theorem 4.3] is algebraic except for the scalar Hardy inequality. Replacing its three-dimensional β by M , the matrices $iM\alpha_j$ are Hermitian and satisfy the same Clifford relations as the α_j . Hence the paper's key matrix $B(x)$ still obeys

$$B(x)^2 = \left| -\nabla \delta(x) + \delta(x) \frac{\nabla \ell(x)}{\ell(x)} \right|^2 \mathbf{1}.$$

Also, on compactly supported spinors in every dimension,

$$\|D_0 \phi\|_2^2 = \langle \phi, -\Delta \phi \rangle = \|\nabla \phi\|_2^2,$$

so the same weak Hardy bound with coefficient $1/4$ enters. Every subsequent estimate in the proof of [1, Theorem 4.3] is scalar. It therefore gives $|\lambda| > (1 + \mu)/2$, the domain statement, and, using the convex-domain Hardy remainder, the endpoint $|\lambda| = 1/2$ without any dimension restriction.

It remains to prove sharpness. On B^d , the standard spinor spherical decomposition [2, Chapter 4] (the same partial-wave mechanism cited in [1, Section 5]) unitarily decomposes the operator into radial first-order systems. After the usual multiplication by $r^{(d-1)/2}$, every channel has, near $r = 1$, the form

$$-iJ \frac{d}{dr} + \frac{\lambda}{1 - r} K + R_\kappa(r),$$

where J, K are anticommuting Hermitian matrices and R_κ is bounded near $r = 1$; the angular eigenvalue contributes only a multiple of $1/r$. Writing $t = 1 - r$, the residue matrix is unitarily equivalent to $i\lambda\alpha_r M$ and has eigenvalues $\pm\lambda$. Standard regular-singular ODE theory therefore gives a fundamental pair with sizes $t^{|\lambda|}$ and $t^{-|\lambda|}$, up to bounded invertible factors (and harmless logarithms in resonant cases). Both are in $L^2(0, \varepsilon; dt)$ exactly when $|\lambda| < 1/2$. The outer endpoint is then limit-circle, so a radial channel, and hence the minimal ball operator, is not essentially self-adjoint. The already proved convex-domain result supplies essential self-adjointness for $|\lambda| \geq 1/2$. $\square$

7 Odd-dimensional mass rotation

Proposition 4 (Exact gauge formula). *Let d be odd, $\rho, \theta \in C^1(\Omega; \mathbb{R})$, and*

$$V = \rho(\cos \theta \beta + \sin \theta \tilde{\beta}), \quad U_\theta = e^{-i\theta \Gamma_d/2}.$$

Then U_θ commutes with every α_j , $U_\theta \beta U_\theta^ = \cos \theta \beta + \sin \theta \tilde{\beta}$, and*

$$U_\theta^* (D_0 + V) U_\theta = D_0 + \rho \beta - \frac{1}{2} \alpha(\nabla \theta) \Gamma_d. \quad (4)$$

In particular, bounded $\nabla \theta$ does not change essential self-adjointness; more singular rotations are covered directly by Theorem 2 when its scalar bound holds.

Proof. For odd d , Γ_d commutes with the kinetic matrices and anticommutes with β . Expanding the exponential on the two eigenspaces of Γ_d gives the rotation identity. The product rule gives $U_\theta^* \partial_j U_\theta = -i(\partial_j \theta) \Gamma_d/2$, which yields (4). If $\nabla \theta$ is bounded, the last term is a bounded Hermitian perturbation. $\square$

8 Verification, novelty, and limitations

The script `code/verify_clifford_classification.py` constructs the Jordan–Wigner matrices, solves the complex matrix anticommutation system for $d = 1, \dots, 8$, and obtains dimensions 2, 1, 2, 1, 2, 1, 2, 1, exactly as predicted. It checks all mass-frame relations, random mass-vector square identities (worst residual 2.802×10^{-17}), and the normal residue eigenvalues for five values of λ in every tested dimension. The full command and output are in `code/verifier_report.txt`. These checks are not used as proof.

A bounded official-arXiv search through 11 August 2026 used arXiv id 2010.09816, the exact phrases “scalar potentials” and “Clifford algebra”, and combinations of “Dirac confinement”, “all dimensions”, and “essential self-adjointness”. It found the source paper but no later exact answer to Comment 5 or the dimension-free threshold question. Novelty still requires expert review.

The theorem concerns the natural minimal Hamiltonian spinor representation specified above. Direct sums with extra multiplicity have larger matrix-valued commutants and require the corresponding multiplicity-space version of the classification. The packet does not solve the separate electric-potential or generic higher-dimensional magnetic-confinement problems in Comments 6 and 8. Human review should focus on the radial-channel limit-point argument and on matching the source’s domain conventions at the critical endpoint.

References

- [1] G. Nenciu, I. Nenciu, and R. Obermeyer, *Essential self-adjointness of symmetric first-order differential systems and confinement of Dirac particles on bounded domains in $\mathbb{R}^d$* , arXiv:2010.09816v2, 2021.
- [2] B. Thaller, *The Dirac Equation*, Texts and Monographs in Physics, Springer, 1992. See Chapter 4 for partial-wave decompositions.